Monte Carlo Integration

Statistical Computing, STA3005

April 20, 2026

Last chapter: Permutation Test

- Goal: Conduct hypothesis testing between two groups of samples without requiring distribution assumption
- Define a test statistics and then approximate the null distribution by permuting or shuffling all of the samples
- The empirical p-value based on B random permutations

$$\hat{p} = \frac{1}{B}\sum_{b=1}^{B} I(T_b^* \geq T_{obs}).$$

- Two-sample Kolmogorov-Smirnov statistic that measures the difference between two distributions

Part I: *Monte Carlo Integration*

Monte Carlo integration is a statistical method based on random sampling. The main target is to numerically compute

$$\int_a^b g(x)dx,$$

especially when the integral doesn't have a closed form.

Intuition

If X is a random variable with density $f(x)$, then the mathematical expectation of the random variable $Y = g(X)$ is

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx.$$

If $x_1, \ldots, x_m$ are i.i.d. draw from $f(x)$, then

$$\hat{\theta} = \frac{1}{m}\sum_{i=1}^{m} g(x_i)$$

is an unbiased estimator of $E[g(X)]$.

Simple Monte Carlo Estimator

Consider the problem of estimating $\theta = \int_0^1 g(x)dx$. Let $x_1, \ldots, x_m \sim \text{Uniform}(0,1)$, then by Strong Law of Large Numbers,

$$\hat{\theta} = \frac{1}{m}\sum_{i=1}^{m} g(X_i) \to \theta$$

in probability.

The simple Monte Carlo estimator of $\int_0^1 g(x)dx$ is $\frac{1}{m}\sum_{i=1}^{m} g(X_i)$.

Example

Compute a Monte Carlo estimate of $\theta = \int_0^1 e^{-x}dx$ and compare the estimate with the exact value.

```
set.seed(123)
m <- 10000
x <- runif(m)
theta_hat <- mean(exp(-x))
print(theta_hat)
```

```
## [1] 0.6333057
```

```
print(1 - exp(-1))
```

```
## [1] 0.6321206
```

Compute an integral on a bounded interval $[a, b]$

Two strategies:

- Change of variables: let $y = (t-a)/(b-a)$ and $dy = (1/(b-a))dt$ s.t.

$$\int_a^b g(t)dt = \int_0^1 g(y(b-a)+a)(b-a)dy.$$

An unbised estimator of $\int_a^b g(t)dt$ is

$$\frac{1}{m}\sum_{i=1}^{m}(b-a)g(y_i(b-a)+a)$$

for $y_1, y_2, \cdots, y_m \sim \text{Unif}(0,1)$.

- Replace $\text{Unif}(0,1)$ by $\text{Unif}(a,b)$.

$$\int_a^b g(t)dt = \int_a^b g(t)(b-a)\frac{1}{b-a}dt = E((b-a)g(X)), X \sim \text{Unif}(a,b)$$

An unbised estimator of $\int_a^b g(t)dt$ is

$$\frac{1}{m}\sum_{i=1}^{m}(b-a)g(x_i)$$

for $x_1, x_2, \cdots, x_m \sim \text{Unif}(a, b)$.

Example

Compute a Monte Carlo estimate of $\theta = \int_2^4 e^{-x}dx$ and compare the estimate with the exact value of the integral.

```
m <- 10000
x <- runif(m, min=2, max=4)
theta.hat <- mean(exp(-x)) * 2
print(theta.hat)
```

```
## [1] 0.1177839
```

```
print(exp(-2) - exp(-4))
```

```
## [1] 0.1170196
```

Compute an integral on unbounded intervals

Example

Use the Monte Carlo approach to estimate the standard normal cdf $\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}dt$.

Notice that the limits of integration cover an unbounded interval, so we use the symmetry of the normal density, that is, $\Phi(0) = \int_{-\infty}^{0} e^{-t^2/2}dt = 1/2$. When $x \geq 0$, we estimate $\theta = \int_0^x e^{-t^2/2}dt$ by Monte Carlo simulation instead.

By making the substitution $u = t/x$, we have

$$\theta = \int_0^1 xe^{-(xu)^2/2}du = E_U[xe^{-(xU)^2/2}], U \sim \text{Unif}(0, 1)$$

Steps of Monte Carlo simulation:

- Generate iid random numbers $u_1, \ldots, u_m \sim \text{Unif}(0, 1)$
- Compute $\hat{\theta} = \frac{1}{m}\sum_{i=1}^{m} xe^{-(xu_i)^2/2}$.
- If $x > 0$, the estimate of $\Phi(x) = 0.5 + \theta/\sqrt{2\pi}$.
- If $x < 0$ compute $\Phi(x) = 1 - \Phi(-x)$.

```
x <- seq(.1, 2.5, length = 10)
m <- 10000
u <- runif(m)
cdf <- numeric(length(x))
for (i in 1:length(x)) {
```

```
  g <- x[i] * exp(-(u * x[i])^2 / 2)
  cdf[i] <- mean(g) / sqrt(2 * pi) + 0.5
}

Phi <- pnorm(x)
print(round(rbind(x, cdf, Phi), 3))
```

```
##     [,1]  [,2]  [,3]  [,4]  [,5]  [,6]  [,7]  [,8]  [,9] [,10]
## x   0.10 0.367 0.633 0.900 1.167 1.433 1.700 1.967 2.233 2.500
## cdf 0.54 0.643 0.737 0.816 0.878 0.924 0.955 0.975 0.987 0.993
## Phi 0.54 0.643 0.737 0.816 0.878 0.924 0.955 0.975 0.987 0.994
```

The Monte Carlo estimates appear to be very close to the `pnorm` values.

As the integrand $\frac{1}{\sqrt{2\pi}}e^{-t^2/2}$ is itself a density function, and we can generate random variables from its density. In particular

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}}e^{-t^2/2}dt = \int_{-\infty}^{\infty} I(t \leq x)\frac{1}{\sqrt{2\pi}}e^{-t^2/2}dt = E[I(Z \leq x)], Z \sim N(0,1)$$

Steps of Monte Carlo simulation:

- Generate random numbers $z_1, \ldots, z_m \sim N(0,1)$.
- Compute the sample mean $\widehat{\Phi}(x) = \frac{1}{m}\sum_{i=1}^{m} I(z_i \leq x)$

```
x <- seq(.1, 2.5, length = 10)
m <- 10000
z <- rnorm(m)
dim(x) <- length(x)
p <- apply(x, MARGIN = 1,
           FUN = function(x, z) {mean(z < x)}, z = z)
Phi <- pnorm(x)
print(round(rbind(x, p, Phi), 3))
```

```
##     [,1]  [,2]  [,3]  [,4]  [,5]  [,6]  [,7]  [,8]  [,9] [,10]
## x   0.10 0.367 0.633 0.900 1.167 1.433 1.700 1.967 2.233 2.500
## p   0.55 0.652 0.743 0.820 0.878 0.922 0.955 0.976 0.987 0.994
## Phi 0.54 0.643 0.737 0.816 0.878 0.924 0.955 0.975 0.987 0.994
```

Compared with `pnorm` function, we have better agreement with `pnorm` in the upper tail, but worse agreement near the center.

General procedure

If $f(x)$ is a probability density function supported on a set A to estimate the integral

$$\theta = \int_A g(x)f(x)dx,$$

- Generate a random sample $x_1, \ldots, x_m$ from the distribution $f(x)$

- Compute the sample mean

$$\hat{\theta} = \frac{1}{m}\sum_{i=1}^{m} g(x_i).$$

- With probability one, $\hat{\theta}$ converges to $E[\hat{\theta}] = \theta$ as $m \to \infty$.

The standard error of $\hat{\theta} = \frac{1}{m}\sum_{i=1}^{m} g(x_i)$.

As $x_1, x_2, \cdots, x_m$ are i.i.d. samples from $f(x)$, then

$$Var(\hat{\theta}) = \frac{1}{m} Var_f(g(X))$$

When the distribution of X is unknown, we use the empirical distribution F_m. The variance of $\hat{\theta}$ can be estimated by

$$\frac{\hat{\sigma}^2}{m} = \frac{1}{m^2}\sum_{i=1}^{m}[g(x_i) - \hat{\theta}]^2.$$

where $\hat{\sigma}^2$ represents the the plug-in estimate of $Var_f(g(X))$.

The corresponding estimate of standard error of $\hat{\theta}$ is

$$\widehat{se}(\hat{\theta}) = \frac{\hat{\sigma}}{\sqrt{m}} = \frac{1}{m}\left\{\sum_{i=1}^{m}[g(x_i) - \hat{\theta}]^2\right\}^{1/2}.$$

- The Central Limit Theorem implies that $\frac{\hat{\theta} - E[\hat{\theta}]}{\sqrt{Var(\hat{\theta})}} \to N(0,1)$ in distribution as $m \to \infty$.
- If m is sufficiently large, $\hat{\theta}$ is approximately normal with mean θ
- Use $\widehat{se}(\hat{\theta})$ to build confidence limits on $\hat{\theta}$

Example

Estimate the variance of the estimator $\hat{\Phi}(x)$, and construct approximate 95% confidence intervals for the estimate of $\Phi(2)$.

```
x <- 2
m <- 10000

out <- matrix(NA, nrow = 4, ncol = 2)
rownames(out) <- c("Estimate", "SE" ,"Lower", "Upper")
colnames(out) <- c("Uniform","Normal")

# Method 1: Generate uniform random number
u <- runif(m)
g_unif <- x * exp(-(u * x)^2 / 2)
v_unif <- mean((g_unif - mean(g_unif))^2) / m

cdf_unif <- mean(g_unif) / sqrt(2 * pi) + 0.5

out[,1] <- c(cdf_unif, sqrt(v_unif/2/pi),
```

```
              cdf_unif - 1.96 * sqrt(v_unif),
              cdf_unif + 1.96 * sqrt(v_unif))

# Method 2: Generate normal random number
z <- rnorm(m)
g_norm <- (z < x) #the indicator function
v_norm <- mean((g_norm - mean(g_norm))^2) / m
cdf_norm <- mean(g_norm)

out[,2] <- c(cdf_norm, sqrt(v_norm),
              cdf_unif - 1.96 * sqrt(v_norm),
              cdf_unif + 1.96 * sqrt(v_norm))

round(out,5)
```

```
##          Uniform  Normal
## Estimate 0.97584 0.97770
## SE       0.00231 0.00148
## Lower    0.96448 0.97294
## Upper    0.98720 0.97873
```

- The SE of $\hat{\Phi}(2)$ using a uniform sample is larger than that using a normal sample
- In the second method, the distribution of $g(X_i) = I(X_i < x)$ is a Bernoulli distribution with successful probability $\Phi(x)$. The theoretical value of $se(\hat{\theta})$ is $\sqrt{0.978(1-0.978)/m} \approx 0.00147$, which is close to the MC estimate 0.00148.

Efficiency

If $\hat{\theta}_1$ and $\hat{\theta}_2$ are two estimators for θ, then $\hat{\theta}_1$ is more efficient (in a statistical sense) than $\hat{\theta}_2$ if

$$\frac{Var(\hat{\theta}_1)}{Var(\hat{\theta}_2)} < 1.$$

- If the variances of estimators $\hat{\theta}_i$ are unknown, we can estimate efficiency by substituting a sample estimate of the variance for each estimator.
- As variance can always be reduced by increasing the number of replicates, we usually do the comparison when the number of replicates for two estimators are the same or two estimators use the same computational time

Part II: *Importance Sampling*

Limitations of simple Monte Carlo method

Recall simple Monte Carlo method

- Generate a large number of replicates $x_1, \ldots, x_m$ uniformly distributed on $[a, b]$
- Estimate $\int_a^b g(x)dx$ by the sample mean $\frac{b-a}{m}\sum_{i=1}^m g(X_i)$

Limitations:

- Not applicable for integral on unbounded intervals
- Be inefficient (large variance) if the function $g(x)$ is far away from a uniform function

Importance sampling

To estimate $\int_{-\infty}^{\infty} g(x)dx$

- Find a density function $\phi(x) > 0$ on the set $\{x : g(x) > 0\}$ and it is easy to sample from $\phi(x)$
- Generate i.i.d. random numbers $x_1, x_2, \cdots, x_m \sim \phi(x)$
- The Monte Carlo estimates by importance sampling is given by

$$\frac{1}{m}\sum_{i=1}^{m}\frac{g(x_i)}{\phi(x_i)}$$

That is because

$$\int g(x)dx = \int \frac{g(x)}{\phi(x)}\phi(x)dx = E[g(X)/\phi(X)], X \sim \phi(x).$$

The variance of the estimator based on $Y = g(X)/\phi(X)$ is $Var(Y)/m$, so the variance of Y should be small to reduce the variance of Monte Carlo integration.

- $\phi(x)$ is called the **envelope** or the **importance sampling function**.
- The variance of Y is small if Y is nearly constant, so the density $\phi(x)$ should be "close" to $g(x)$.

Example

We consider several possible choices of importance functions to estimate

$$\int_0^1 \frac{e^{-x}}{1+x^2}dx.$$

The candidates for the importance functions are

- Uniform distribution: $\phi_0(x) = 1, \quad 0 < x < 1$
- Exponential distribution: $\phi_1(x) = e^{-x}, \quad 0 < x < \infty$
- Cauchy distribution: $\phi_2(x) = \frac{1}{1+x^2}/\pi, \quad -\infty < x < \infty$
- Truncated exponential distribution: $\phi_3(x) = e^{-x}/(1-e^{-1}), \quad 0 < x < 1$
- Truncated Cauchy distribution: $\phi_4(x) = \frac{4}{\pi(1+x^2)}, \quad 0 < x < 1$

```
m <- 10000
theta.hat <- rep(NA, 5)
se <- rep(NA, 5)

g <- function(x) {
  exp(-x - log(1+x^2)) * (x > 0) * (x < 1)
}

# Using phi0
```

```r
x <- runif(m)
fg <- g(x)
theta.hat[1] <- mean(fg)
se[1] <- sd(fg)

# Using phi1
x <- rexp(m, 1)
fg <- g(x) / exp(-x)
theta.hat[2] <- mean(fg)
se[2] <- sd(fg)

# Using phi2
x <- rcauchy(m)
i <- c(which(x > 1), which(x < 0))
x[i] <- 2 #to catch overflow errors in g(x)
fg <- g(x) / dcauchy(x)
theta.hat[3] <- mean(fg)

# Using phi3, inverse transform method
se[3] <- sd(fg)
u <- runif(m)
x <- - log(1 - u * (1 - exp(-1)))
fg <- g(x) / (exp(-x) / (1 - exp(-1)))
theta.hat[4] <- mean(fg)
se[4] <- sd(fg)

#Using phi4, inverse transform method
u <- runif(m)
x <- tan(pi * u / 4)
fg <- g(x) / (4 / ((1 + x^2) * pi))
theta.hat[5] <- mean(fg)
se[5] <- sd(fg)

round(rbind(theta.hat, se/sqrt(m)),5)
```

```
##              [,1]    [,2]    [,3]    [,4]    [,5]
## theta.hat 0.52178 0.53017 0.52760 0.52510 0.52373
##           0.00244 0.00416 0.00954 0.00097 0.00141
```

- ϕ_1 and ϕ_2 have larger ranges and many of the simulated values will contribute zeros to the sum, so they are inefficient
- ϕ_3 and ϕ_4 are truncated distribution to restrict random numbers generated on $(0, 1)$.
- Compared with uniform distribution ϕ_0, their densities are closer to integrand, so they are more efficient than ϕ_0

Variance in Importance Sampling

If $\phi(x)$ is the importance sampling distribution, and X has pdf $\phi(x)$ supported on A, then

$$\theta = \int_A g(x)dx = \int_A \left[\frac{g(x)}{\phi(x)}\right] \phi(x)dx = E\left[\frac{g(X)}{\phi(X)}\right].$$

If $x_1, \ldots, x_n$ is a random sample from $\phi(x)$, the estimator is

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^{n} \left[\frac{g(x_i)}{\phi(x_i)} \right].$$

The variance of the estimator is

$$\mathrm{Var}(\hat{\theta}) = E[\hat{\theta}^2] - (E[\hat{\theta}])^2 = \frac{1}{n} \left[\int_A \left(\frac{g^2(x)}{\phi(x)} \right) dx - \theta^2 \right].$$

By Cauchy-Schwarz Inequality,

$$\int_A \left[\frac{g^2(x)}{\phi(x)} \right] dx \int_A \phi(x) dx \geq \left(\int_A |g(x)| dx \right)^2$$

Equality holds if and only if $\phi(x) \propto \frac{g(x)^2}{\phi(x)}$, that is,

$$\phi(x) = \frac{|g(x)|}{\int_A |g(x)| dx}$$

Thus, the minimum variance is $\frac{1}{n} \left[\left(\int_A |g(x)| dx \right)^2 - \theta^2 \right]$

- However, the problem is to estimate $\int_A g(x) dx$, so the value of $\int_A |g(x)| dx$ in the denominator of $\phi(x)$ is not available.
- Although it may be difficult to choose $\phi(x)$ to attain minimum variance, variance may be "close" to optimal if $\phi(x)$ is chosen so that the shape of the density $\phi(x)$ is "close to" $|g(x)|$ on A.
- Be careful to select an importance function that results in small variance of Monte Carlo integration

Summary

- Goal: Estimate $\int_A g(x) f(x) dx$ by random sampling, where $f(x)$ is a PDF.
- Monte Carlo method
  1. Generate a large number of replicates $x_1, \ldots, x_m \sim f(x)$
  2. Estimate $\int_A g(x) f(x) dx$ by the sample mean $\frac{1}{m} \sum_{i=1}^{m} g(x_i)$
- When A is a bounded interval $[a, b]$, simply select $f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$
- To reduce the variance of the estimator, importance sampling suggests that $\phi(x)$ is easy to sample from and "close" to $g(x) f(x)$